Artificial Infringement

Michael P. Goodyear
New York University School of Law
New York, NY, USA
michael.goodyear@nyu.edu

ABSTRACT

By examining the historical and doctrinal response of copyright law to new technologies, this Article offers a new analytical framework for determining liability for what it terms artificial infringement, or infringing outputs created by generative AI systems. Time and again, new technologies have posed challenges to existing copyright law, straining its capacity to balance protecting authors' rights to incentivize new expression and providing public access to their works. Courts and Congress have been able to maintain this balance by using a variety of doctrinal tools, including fair use, compulsory licensing, and secondary liability. One underexamined tool, however, is the refinement of direct liability. This Article reveals how courts introduced the causation requirement to maintain copyright's balance in response to complex machine-generated infringements.

Together, direct liability's causation requirement and other doctrinal tools provide a viable framework for maintaining copyright's incentives-access balance despite the acute challenges of artificial infringement. By holding the AI system directly liable, courts can utilize and refine secondary liability doctrines to conduct a more nuanced analysis of user and developer liability for AI-generated infringements. Along with fair use, these refinements to copyright doctrine provide a more comprehensive resolution to the battles between copyright and AI.

ACM Reference format:

Michael P. Goodyear. 2025. Artificial Infringement. In *Proceedings of ACM Symposium on Computer Science and Law (CSLaw '25), March 25–27, 2025, München, Germany*. ACM, New York, NY, USA, 12 pages.

1 INTRODUCTION

Copyright is one of the leading legal and policy battlegrounds for generative AI. So far, debate has centered on the question of whether training an AI system is lawful under copyright law. Several scholars have convincingly posited that using copyrighted works for AI systems' training data is a permitted fair use [Sag 2023] [Henderson et al. 2023b] [Samuelson 2023] [Lee et al. 2025] [Bracha 2023]. Over three dozen pending U.S. court cases grapple with this question of whether the training of generative AI systems on copyrighted works is lawful or infringement.

Yet even if training an AI system on copyrighted works is permitted under copyright law, there is still the question of who should be liable when a generative AI system creates an infringing output. Even the staunchest advocates of fair use have admitted that in at least a few cases, some outputs would likely infringe copyright because they are substantially similar to works in the training data and compete in the same market as the original copyrighted work [Henderson et al. 2023b] [Sag 2024]. This generative aspect of AI systems is novel and troubling under copyright law. Overbroad liability could risk stifling the generative AI industry and its potential innovative advances. Holding the developer or user de facto liable for any resulting infringements, despite their best efforts, is misaligned with copyright precedent and harms incentives to build and use AI. Yet depriving authors of their full rights under copyright law could harm the creativity the Copyright Act seeks to foster. As others have recognized, AI may necessitate a new legal paradigm [Abraham and Rabin 2019] [Beckers and Teubner 2023] [Price and Cohen 2024] [Selbst 2019].

While approaches to AI-generated harms have started to ossify in the broader tort context, especially for autonomous vehicles, discussion of what this Article terms "artificial infringement"—infringing outputs created by generative AI systems—is still nascent. In one rare example, Rosati [2024] offered an overview of potential output liability issues from an EU and UK perspective. Meanwhile, the main U.S. exception centers on the question of "volition" in copyright law. Chatterjee and Fromer [2019] proposed that AI systems might, under the right circumstances, be sufficiently volitional within the meaning of copyright law for the developer to be held liable because the doctrine focuses on function rather than consciousness. They argued that while a copy machine would not have functional volition, "a more sophisticated computer could plausibly be designed to 'choose' whether to make a copy."

This Article takes a different approach, relying not on volition but on causation to propose that AI systems themselves should be directly liable for infringing outputs because they cause them to occur. To make this contribution, it looks to the history of copyright's direct liability doctrine. Courts and Congress have used a variety of tools to accommodate new technologies while maintaining a balance between copyright's goals of protecting copyright owners' rights and providing public access to their works. However, the refinement of the direct liability test to include causation has been neglected as a tool in the literature. Situating AI

Permission to make digital or hard copies of part or all of this work for personal or classroom use is granted without fee provided that copies are not made or distributed for profit or commercial advantage and that copies bear this notice and the full citation on the first page. Copyrights for third-party components of this work must be honored. For all other uses, contact the owner/author(s).
CSLaw '25, March 25–27, München, Germany.
© 2025
ACM ISBN 979-8-4007-1421-4/25/03.
https://doi.org/10.1145/3709025.3712208

within this narrative demonstrates that, as a doctrinal matter, the AI system is often the cause of infringement because the infringing quality of the output is attributable to it.

Holding the AI system directly liable would shift developer and user liability to a more nuanced, intent-focused secondary liability framework. Furthermore, holding the developer or user directly liable for all artificial infringements would punish them for offering or using a semiautonomous generative AI system, even though they have limited control over the specific output. Instead, recognizing the AI system as a direct infringer allows the analysis to proceed to the more flexible, fault-based contributory liability and inducement standards for determining when to hold developers or users liable for their roles in the resulting infringement. Expanding beyond the fair use analysis that has been offered so far, this Article's approach uses two tools in tandem—refinements to direct and secondary liability—to maintain copyright's incentives-access balance in the face of new challenges posed by artificial infringement.

2 MAINTAINING COPYRIGHT'S BALANCE

New technologies have long posed serious questions for copyright law. One of the latest innovations, generative AI, is no different. Already, scholars and litigants have debated whether training generative AI models on copyrighted materials should be lawful or not. However, there is another copyright liability question that has been largely unexamined in the literature: when an AI system generates an infringing copy, who should be liable? The unpredictability of AI-generated outputs makes this a case of first impression for copyright law. Courts must proceed carefully, as an improper liability framework could risk upsetting the delicate balance copyright law seeks to maintain between its competing goals of protecting copyright owners' rights to incentivize new creative works and providing public access to those works.

2.1 The Incomplete Fair Use Defense

There are three sequential questions that courts must address in copyright infringement litigation involving generative AI to determine liability. First, is the training of a generative AI system on copyrighted material fair use or otherwise noninfringing? Second, if so, is a specific output from a generative AI system fair use or otherwise noninfringing? Finally, if the output is infringing, *who* is held liable and under what circumstances?

So far, scholars and litigants have largely restricted themselves to examining the first two questions, largely focused on fair use. Fair use is a statutory exception to copyright infringement for certain socially-beneficial uses such as criticism, education, and other transformative purposes. Courts weigh four non-exclusive factors as a sort of proxy for differentiating between uses we wish to encourage and those we would rather prohibit in favor of users' rights [Copyright Law of the United States 1992].

There are at least two opportunities for a copyright infringement claim and fair use defense with generative AI: at the input (training) stage, and at the output stage. The first question of the analysis is input-side fair use, whether training a generative AI model on copyrighted works infringes the reproduction or derivative works right—or is a lawful fair use. During the training process, temporary or incidental copies of the data may be created, which could result in a prima facie infringement claim. While some of the millions of works on which these models are trained are in the public domain or otherwise free to use, many are also copyrightable [Guadamuz 2024]. This is the question U.S. courts are facing in the plethora of generative AI-related lawsuits.

Several scholars have already weighed in on this question of input infringement, offering robust fair use or other non-infringement arguments. Lemley and Casey [2021], Levendowski [2018], Sag [2023], and Samuelson [2024] argue that training an AI system on copyrighted content is likely transformative. These arguments place generative AI inside a long line of functionally transformative devices and systems that courts have deemed to be fair uses, including search engines [Perfect 10, Inc. v. Amazon.com, Inc. 2007] [Kelly v. Arriba Soft Corp. 2003], plagiarism checkers [A.V. ex rel. Vanderhye v. iParadigms, LLC 2009], and Google Books [Author's Guild v. Google, Inc. 2015]. Alternatively, Oren Bracha [2023] argues that AI training relies on the uncopyrightable parts of copyrighted works, such as their underlying facts or grammatical structure, and therefore should not implicate copyright law.

While these input-side fair use and non-infringement arguments are persuasive, even if they were to succeed, we would still have the second question: whether a particular output is infringing. An AI-generated output can be a copy, and therefore implicate copyright law, because it is fixed in a computer's random-access memory ("RAM") for more than a transitory duration [MAI System Corp. v. Peak Computer, Inc. 1993]. Unlike the input analysis, whether an output is noninfringing will likely require individualized examinations of each output. In many cases, the output will be noninfringing or fair use will likely apply.

AI systems likely engage in actual copying, which is required for copyright infringement, at least some of the time. Copyright infringement requires copying in fact and improper appropriation [Rentmeester v. Nike, Inc. 2018]. While substantial similarity would be determined on a case-by-case basis, training the AI system on the original work could support copying in fact. Cooper and Grimmelmann [2025] have suggested that if an AI model "memorizes its training data, the training data is *in the model*." Regurgitation is copying and implies memorization. Even if we were to reject Cooper and Grimmelmann's argument, exposure to a work can be sufficient to show copying because even subconscious copying counts. For example, in *Three Boys Music Corp. v. Bolton* [2000], the court found that the defendants may have listened to Plaintiff's song, "Love Is a Wonderful Thing," twenty years prior as teenagers and copied it from the depths of their memories. Similarly, if the AI system was trained on a particular work, it is possible it is copying it in fact and that a substantially similar output is infringing.

Infringing AI-generated outputs are certain to occur in at least some circumstances. Although the creators of AI systems can—and should—do their best to restrict infringement, the unpredictability of AI outputs mean that infringements are bound to occur

[Marchisio 2021]. Many scholars and studies recognize that at least a portion of outputs may be substantially similar to copyrighted works but not fall within the confines of the fair use doctrine [Lemley & Casey 2021] [Henderson et al. 2023b] [Lee et al. 2025] [Samuelson 2024] [Carlini 2023] [CopyrightCatcher 2024]. Infringement is especially likely with copyrighted works that are more abstractly protected, such as characters [Sag 2023] [Lee and Grimmelmann 2024].

There is also the issue of prompt injection or adversarial machine learning, through which individuals can make machine learning systems ignore instructions and act in unintended ways [Calo et al. 2018]. Sag [2023], for example, successfully provoked Midjourney and Stable Diffusion into generating likely infringing images of Peanuts' Snoopy, Disney's Mickey Mouse, and Banksy's *Girl with Balloon*. Similarly, Lee and Grimmelmann [2024] made Stable Diffusion and GPT-4 produce images that are analogous to Nintendo's cartoon Italian plumber Mario.

While these scholars intended to make the AI systems generate infringing outputs, infringements may also occur where they are not intended. For example, in Sag's [2023] work, one of the prompts that generated an output likely infringing on Banksy's *Girl with Balloon* was "black and white stencil little girl reaching for heart-shaped red balloon." It is foreseeable that others may use a similar prompt without thinking of or even being aware of Banksy's work. Nonetheless, the AI system may generate an infringement like the one Sag identified, and someone will be held responsible.

Therefore, generative AI raises an important but largely unexamined third question: who is liable when generative AI inevitably creates substantially similar, infringing outputs? The unpredictability of AI outputs makes this liability question a novel one for copyright law. There are no reliable techniques for researchers or users to always steer large machine learning-based AI systems away from undesired outcomes [Bowman 2023]. Henderson et al. [2023b] suggested that machine learning researchers should ensure that AI systems create transformative outputs that would likely be fair use. But even when various precautions are (rightly) implemented in advance, developers can still not predict machine learning outputs with complete accuracy [Lemley and Casey 2019]. It is hard or even impossible to establish foolproof guardrails to prevent AI systems from ever generating infringing outputs [Bowman 2023].

2.2 Balancing Methods

Proper answers to this new liability question must fit within copyright's existing paradigm of balance between incentives and access. The U.S. Constitution [1789] enshrines copyright's goals of encouraging creation of new expressive works and the dissemination thereof. Copyright gives authors of original, fixed expressive works a set of exclusive rights—including the right to copy, make derivatives, and publicly display and perform that copyrighted work—to incentivize them to create [Copyright Law of the United States 1990]. But granting unfettered rights would harm copyright's other goal, the "dissemination of [said] creative works" [Stewart v. Abend 1990].

Therefore, as the Supreme Court has noted, copyright law seeks "a balance" between protecting a copyright holder's rights to incentivize creation and providing public access to their works [Sony Corporation of America v. Universal City Studios, Inc. 1984]. Scholars have long strived to preserve this balance too [Samuelson et al. 2010] [Ginsburg 2002]. Maintaining this balance between rights and access is essential to achieving the Constitution's mandate to "promote the progress of knowledge and learning" [Loren 2003].

New technologies pose a particularly thorny challenge to the incentives-access balance, placing an onus on courts and Congress to maintain it. As one federal judge put it, new technologies lead to a "push and pull . . . over the breadth or narrowness" of copyright law [Karnow 2000].

Although the incentives-access balance is universally desired, the optimal balance is difficult to ascertain. The law uses a toolkit of imperfect proxies and polestars to tease out what threatens and fosters the incentives-access balance in order to maintain it.

The most prominent example is fair use, often considered the most important exception to copyright infringement. But while fair use has been called the "*de facto* regulator of new media technologies" [Ard 2025], it is not the lone tool courts can use. Other avenues are needed to permit innovation and maintain the incentives-access balance.

Congress and courts have maintained copyright's incentives-access balance in new technology cases primarily through the use of one of five doctrines: (1) fair use, (2) explicit exclusions, (3) licensing, (4) direct liability, and (5) secondary liability. Recent scholarship has sought to explore these other approaches [Goodyear 2025] [Victor 2020].

The largely missing balance approach in the scholarly literature is direct liability. While several scholars have analyzed the so-called "volition or causation" requirement in copyright direct liability [Denicola 2016] [Nimmer 2019] [Chatterjee and Fromer 2019], they do not discuss how this could fit into an overall approach for accommodating new technologies while also maintaining copyright's incentives-access balance. This Article seeks to fill this lacuna by understanding how direct copyright infringement liability responded to technology, deducing a refinement guide from this history, and applying that guide to the emerging challenges of generative AI.

3 CAUSATION AND MECHANICAL INFRINGEMENT

For much of U.S. history, no one was all that concerned with who would be held directly liable for copyright infringement if it occurred. The direct infringer—if there was one—was the corporeal individual who physically carried out the infringement. But the advent of more complicated technologies with automatic functions and multiple actors complicated what I term the "corporeal infringement" standard, leading to the development of a new paradigm, premised on causation, for what this Article calls "mechanical infringement."

3.1 Corporeal Infringement

This Article coins the term corporeal infringement to describe the original, quintessential type of infringement that is carried out by the human hand. While this term was never used by the courts or scholars, this was the reality for much of U.S. copyright history since the eighteenth century prior to the development of more complex machines. An individual reproducing, making a derivative of, distributing, publicly or digital performing, or publicly displaying another's copyrighted work by their own hand, without an exception or defense, is guilty of corporeal infringement [Copyright Law of the United States 2002].

To establish a prima facie claim for direct copyright infringement, a copyright owner must show copying in fact and improper appropriation [Rentmeester v. Nike, Inc. 2018]. Copying in fact requires that the defendant's work is strikingly similar to theirs or is substantially similar in cases where the defendant had access to plaintiff's work [Boisson v. Banian 2001] [Selle v. Gibb 1984]. Improper appropriation requires that sufficient protected expression was copied from the original [Rentmeester v. Nike, Inc. 2018]. Corporeal infringement was not concerned with *who* copied in fact because that was self-evident.

3.2 Mechanical Infringement

The rules of corporeal infringement existed, largely unencumbered, for decades, but technological developments put them under pressure. The advent of more complicated machines caused uncertainty to emerge in the corporeal infringement test. Copying in fact occurred, but the identity of the copier was less clear. When a plethora of actors was involved in the operation of a facially neutral machine that produced infringement, who really caused the infringement to occur?

In response to these new challenges to copyright's incentives-access balance, courts elucidated an additional requirement in the doctrine: "volition or causation," which courts often condense into simply "volition." The term was muddled in the initial decision that introduced it, although that decision focused on actions rather mental state [Religious Technology Center v. Netcom On-Line Communication Services, Inc. 1995]. Scholars and courts have subsequently suggested the term means willfulness, causation, or both [Chatterjee and Fromer 2019] [Hunley v. Instagram, LLC 2023] [Denicola 2016] [Goold 2015]. However, causation seems to be driving the analysis for the increasingly complex machines of this second stage of direct liability, which I term "mechanical" infringement, and I therefore focus on the causation half of the "volition or causation" requirement in this Article.

The "volition or causation" requirement emerged in one of the early cases involving infringement liability over the Internet. In *Religious Technology Center v. Netcom On-Line Communication Services, Inc.* [1995], Plaintiffs sued an Internet service provider for providing services to an online bulletin board that hosted allegedly infringing content. The court acknowledged that (direct) copyright infringement was largely strict liability, limiting his options, but thought there must be a limit in order to maintain copyright's incentives-access balance. Otherwise, claims would run roughshod over tangentially related entities such as Internet service providers that only provided the initial setup for all comers, disincentivizing innovation by aggrandizing the bounds of copyright. The judge thought "there should still be some element of *volition or causation* which is lacking where a defendant's system is merely used to create a copy by a third party. . . . without any human intervention." He held that Netcom was not directly liable because its actions were only "automatic and indiscriminate"—the users were the ones who had caused the infringing copies to be made. But Netcom could still potentially be held liable under secondary liability.

Since *Netcom*, causation has become an established feature of copyright law that has helped maintain the incentives-access balance in mechanical infringement cases. It has served as a rule that creation, supply, or ownership of a machine, by itself, does not make one directly liable. The requirement makes intuitive sense, as it would be undesirable to hold technology providers liable for all infringements resulting from their technologies that are capable of non-infringing, value-adding uses [Chatterjee and Fromer 2019]. For example, in *Cartoon Network, LP, LLLP v. CSC Holdings, Inc.* ("*Cablevision*") [2008], Cartoon Network sued the defendant cable company for allegedly directly infringing copyrighted works by offering its viewers a remote-storage digital video recorder. The court reasoned that while Cablevision had designed and maintained the system, the user was the direct cause because they "actually presse[d] the button to make the recording." Subsequent case law has reaffirmed the causation requirement [BWP Media USA, Inc. v. T & S Software Associates, Inc. 2017] [ABKCO Music, Inc. v. Sagan 2022].

4 ARTIFICIAL INFRINGEMENT

A new technology, generative AI, now brings us into the third era of direct copyright infringement: AI-generated infringement, or "artificial infringement." Computer and Internet systems have always relied on certain automated functions, such as keys causing corresponding actions, files being saved, and webpages responding to user clicks. But generative AI is more than merely pressing a button for a narrow, desired outcome. This type of AI involves two types of code: programmer-created and machine-created code. First, the programmer writes the code for the model. Second, the program then consumes the (programmer-selected) training data and generates its own code that executes outputs in response to specific user inquiries [Jatinder et al. 2016]. While both sets of code are generally observable, generative AI tools are non-deterministic, meaning that the same inputs can produce different outputs [Cooper et al. 2022]. is still semiautonomous and the exact outputs it creates are unpredictable. The exact path the AI system follows can be complex, messy, unexpected, and inexplainable [Desai and Kroll 2017]. While the general functions are desired and predictable, the non-deterministic nature of generative AI systems means specific outputs often cannot be predicted ahead of time, especially with larger generative models [Ganguli et al. 2022]. Unlike with previous technologies that created infringements, such as copy machines or VCRs, the developer and user of generative AI does not know the identity of the infringing work in advance. Breaking

from the tool-like machines of the past, generative AI presents new questions for copyright infringement liability that require their own paradigm to avoid upsetting copyright's incentives-access balance.

The novel features of generative AI make artificial infringement a difficult balancing challenge for copyright law. An overbroad direct liability rule for artificial infringement could discourage robust, competitive generative AI innovation due to the resulting liability risk—up to $150,000 per infringed work [Copyright Law of the United States 2010a]. But too weak a rule could allow infringements to proliferate, discouraging creation of new works. These concerns are salient. Scholars have already struggled to craft new liability rules for AI-caused harms in other areas of law, including vehicular accidents and medical malpractice [Abraham and Rabin 2019] [Vladeck 2014] [Hubbard 2014] [Davola 2018] [Price and Cohen 2024] [Gerke 2021] [Tschider 2021]. This Article seeks to fill the copyright infringement liability gap in the literature and the law by situating artificial infringement within the underexplored historical evolution of the direct infringement test.

4.1 Unpredictable AI-Generated Works

The main complication for determining who is liable for an infringing output is the difficulty of predicting a specific AI-generated output in advance. While AI systems are achieving impressive new heights, their increasing size, complexity, and reliance on machine learning is making their outputs increasingly difficult to predict [Bowman 2023] [Ganguli et al. 2022]. In addition, machine learning often incorporates randomness, which can help achieve the best outcome, but at the cost of accountability and reproducibility [Kroll et al. 2017]. Even AI systems' programmers may not be able to precisely predict (or sometimes even explain) their outputs [Ganguli et al. 2022] [Hugenholtz and Quintais 2021] [Lior 2020]. In addition, the more complex or ill-defined the goal, the less stable, predictable, and explainable the AI system's actions will be. Continuous learning by AI systems exacerbates the problem of humans predicting outputs in advance [Rachum-Twaig 2020].

Yet the unpredictability of an AI system's outputs is a feature, not a fault. It is what enables AI systems to go beyond human capabilities and offer novel outputs [Duffourc 2023]. While a simpler algorithm may have more predictable outputs, such algorithms have more limited functionality. Instead of following a single formula, machine learning allows AI systems to achieve high-level goals whose solutions (or specific outputs) are often unknown in advance by the human operators [Desai and Kroll 2017]. This means that at least a certain degree of unpredictability is inherent in advanced generative AI systems' outputs.

Machine learning only achieves this power by sacrificing control and predictability [Lemley and Casey 2019]. There are no reliable techniques for researchers or users to always steer larger machine learning-based AI systems to a specific desired output [Bowman 2023]. Henderson et al. [2023b] suggested that machine learning researchers should ensure that AI systems create transformative (likely fair use) outputs. Even when various precautions are implemented in advance, it is still difficult to establish foolproof guardrails to prevent AI systems from generating specific outputs [Bowman 2023]. AI models can misinterpret prompts, overrely on specific training material, or suffer from analysis disconnects that could lead to a very different output from what the user intended [Henderson et al. 2023b]. Therefore, the semiautonomous nature of the AI systems means some infringements will be genuinely unforeseeable despite programmers' and users' best efforts [Lemley and Casey 2019]. This output unpredictability is one of AI's greatest strength, but it would also seem to create the very real risk of copyright infringement liability.

4.2 Direct Liability for AI

Copyright law needs a viable artificial infringement framework to fill this new liability gap. Many entities are involved in the creation of the generative AI system that makes the infringing output: the user, the parties involved in the supply chain that creates the AI system (here I refer to this group collectively as the "developer"), and the AI system itself.

Like with mechanical infringement, causation—specifically proximate causation—can provide a valuable framework for assigning liability. Tort law recognizes two types of causation: but-for and proximate. But-for causation means that but for the defendant's action, the injury would not have occurred [Comcast Corp. v. National Association of African American-Owned Media 2020]. Proximate causation means the defendant's conduct was sufficiently related to the plaintiff's injury for liability to attach [Lexmark International, Inc. v. Static Control Components 2014]. Both the developer and the user are but-for causes of the infringing output, as it would not exist without the developer providing the AI system or the user inputting the prompt. This Article instead focuses on proximate causation because it seeks to answer to whom or what we should attribute the infringing nature of a given output.

A proper application of the causation requirement would make the AI system itself the direct infringer because it caused the infringement to occur. This outcome then allows courts to use another copyright balancing tool, secondary liability, to apply a nuanced framework for evaluating developer and user liability. By using these tools together, courts can maintain copyright's incentives-access balance despite the novel issues posed by artificial infringement.

4.2.1 Limited Instances of Potential User and Developer Causation. There are some limited cases where the developer or user could have caused the infringement. Rosati [2024] assumed, at least from an EU and UK perspective, that generally the user would be held directly liable. Rosati may be correct that the user may have caused the infringement when they provide the AI system with a prompt that clearly seeks to generate an infringing output.

User liability has been contemplated in other areas of AI based on a causation analysis. Take, for example, so-called autonomous vehicles, which are perhaps the most explored area of AI liability. These vehicles are not yet truly autonomous and always involve human input to some degree [Abraham and Rabin 2019]. Accordingly, current autonomous vehicle liability law is driver-focused, as the vehicle remains a mere tool in the hands of the

driver. For example, courts have held operators liable for accidents where the vehicle was in autopilot [Brouse v. United States 1949] [Came v. Micou 2005]. If the driver makes a mistake, they are liable—unless a manufacturing defect is present—because they were the cause of the accident [Karney v. Leonard Transportation Corp. 2008]. Yet scholars have posited that drivers of *fully* autonomous vehicles should not be held liable for the harms inflicted by the vehicles, since they would be unable to affect the car's behavior or the underlying AI system. Instead, they explain, the liable party should be one who can affect the functioning of autonomous vehicles—the one who caused the accident to occur [Abraham and Rabin 2019] [Davola 2018].

The developer could also craft an AI model that is highly likely to infringe. Under the mechanical infringement paradigm, the developer could be held directly liable where they devised the system in a way that it was almost guaranteed to infringe when it was used. For example, in *American Broadcasting Corporation v. Aereo* [2014], the provision of all the copied content by Aereo for profit and in competition with the broadcasters that owned the rights to the content was enough to make it directly liable even though the users chose the specific (infringing) content. In *Capitol Records, LLC v. ReDigi Inc.* [2013], the Southern District of New York found that ReDigi "programmed [its] software to choose copyrighted content [(i.e., digital music files), which] satisfie[d] the volitional conduct requirement."

However, providers of automated or hands-off systems have generally not satisfied the volition or causation requirement without more. This was the case in *Netcom* [1995] and *Cablevision* [2008], where the mere supply of Internet services and a remote DVR device did not cause the infringement. Similarly, cloud storage providers and website hosts have not met the causation requirement for users' infringements when they merely provided neutral services that did not treat infringing content any differently [Disney Enterprises, Inc. v. Hotfile Corp. 2011] [Fox Broadcasting Co. v. Dish Network LLC 2014]. This is similar to the *Sony* [1984] rule in secondary liability that shields the provider of a product from constructive knowledge of resulting infringement so long as the product has substantial noninfringing uses. Based on this precedent, the mere provision of an AI system should not make the developer liable, but the developer could be the cause of infringement if it designs the AI system to infringe or its outputs are effectively guaranteed to be infringing.

As machines become more autonomous, the causation line is increasingly hazy. AI systems' opacity and unpredictability make it difficult at best to show causation by the developer or user of the AI system [Lior 2020]. For example, Kaminski [2023] has written that "[w]hen an AI system causes harm, it can be particularly hard to determine causality *ex post*, in part because such systems are often technically and legal opaque. AI harms are often not foreseeable by a human developer, which make them harder to disincentivize through dyadic litigation." This creates a potential problem where infringement occurs but there is no identifiable perpetrator to whom the harm can be attributed [Calo 2015].

With other torts, courts and scholars have started to similarly conclude that existing law does not align with how AI functions. For example, a judge on the San Francisco County Superior Court concluded that it will be difficult to apply traditional tort doctrines to autonomous robots [Karnow 2016]. Balkin [2015] similarly concluded that traditional liability would be a poor fit for self-learning systems because "[c]ode is lawless." Other scholars have concluded that strict or de facto liability for AI systems would lead to unjust outcomes [Rachum-Twaig 2020] [Lior 2020] [Scherer 2018] [Karnow 1996].

4.2.2 The Promise of AI System Causation. This Article proposes that the AI system itself could be directly liable for causing AI-generated infringements. Chatterjee and Fromer [2019] previously proposed, as a philosophical matter, that AI systems might be said to act with volition if they can evaluate whether an output is likely to infringe upon copyright and decline to make such a copy. Here, I instead show, as a historical and doctrinal matter, that AI systems could be considered directly liable under causation because they cause the infringement to occur. While the developer is providing the initial setup and parameters, and the user provides the specific prompt, the AI system is determining the actual expression and the infringing nature of that expression. A causation analysis suggests that making the AI system the direct infringer (but not the developer's agent, as will be discussed below, Section 4.2.4) is the correct outcome. Importantly, a direct infringer is also necessary to reach secondary liability, which offers a non-de facto framework for analyzing the developer and users' liability [A&M Records, Inc. v. Napster, Inc. 2001]. This understanding of the AI system as the direct infringer allows courts to engage in a more nuanced secondary liability framework for determining developer and user liability, which will be discussed below (Section 4.3). Generative AI systems are not mere mechanical tools, like a copy machine, that merely reproduce what its users provide it. Chopra and White [2011] suggested thinking of two scenarios with AI: where the AI system is a tool of the user and where it can be thought of as an independent actor or agent. This paradigm is helpful because it looks to proximate causation of the harm rather than simply seeking a party to remunerate the rights owner. In the autonomous vehicle context, no one contemplates holding the car liable, but this is because it is usually not the sole decision-maker: the driver (or in some cases the developer) is often able to step in and act to prevent a harm from occurring. But generative AI operates semi-autonomously to achieve a human-provided objective in a way that could result in an unpredictable output due to the AI using its own training to reach decisions. There are cases where the AI system is committing torts in irreducible ways where there is not necessarily a consciously responsible person [Abbott & Sarch 2019]. In the copyright context, a human user or developer cannot accurately predict an output or reverse its generation once it has occurred. The developer and user could foresee that an AI system could be used to infringe, much like the manufacturer of a copy machine could foresee that it might be used to make an infringing copy. But copyright law is more concerned with whether the specific infringements are foreseeable than general foreseeability of a device being used to harm, which was rejected in cases like *Sony* [1984].

In many cases, the AI system seems to be the one that proximately causes the infringement because it is making determinations about what and how expressive content should be included in the output. Generative AI has started to shift from mere tool to a semiautonomous system. Even though the user provides the prompt, they and the developer often cannot predict the exact output. The prompt merely sets the parameters of the idea that the AI system will develop with specific expression. The AI system determining the actual expression (including infringing expression) seems to place causation in the hands of the generative AI system itself. Mulligan [2018] previously suggested this possibility in the tort context, noting that with the rise of black-box algorithms, robots could be the cause of the harms. Chatterjee and Fromer [2019] also suggested that AI could be considered to "choose" whether to make a specific (infringing) copy.

As a legal matter too, it would seem that the AI system could be deemed the direct infringer. The Copyright Act seems to contemplate infringement by non-humans since § 501 [2019] says "anybody"—including state governments—can be liable for infringement. The law frequently holds corporations responsible for infringements and other torts [Petroleos Mexicanos Refinacion v. M/T KING A 2009] [Miller 2021]. Jurisdictions have been willing to recognize new types of non-human entities that could be liable too, such as Wyoming accepting decentralized autonomous organizations ("DAOs") [Corporations Law of Wyoming 2023].Congress could mandate that AI systems are the direct infringers in a statute, but courts also have some flexibility to accept this conception by themselves. After all, Judge Whyte introduced the "volition or causation" requirement for mechanical infringement *sua sponte* [Religious Technology Center v. Netcom On-Line Communication Services, Inc. 1995].

4.2.3 Normative Detriments of De Facto Liability for Developers and Users. As a policy matter, there should not be a general rule that would hold AI developers or users directly liable for all infringing outputs. Such a rule would be overreaching and unmoored from causation considerations because: (1) it conflicts with precedent on algorithmic functions, (2) the AI system, not a human, typically selects the (infringing) expression, (3) it could chill AI innovation and competition, and (4) it would not advance the traditional tort goal of deterrence.

First, a per se liability rule for the developer or user would likely conflict with precedent. Previous copyright cases held that service providers were not directly liable for automatic functions without more [Parker v. Google, Inc. 2006] [Field v. Google Inc. 2006]. Recently, the U.S. Supreme Court also held in *Twitter, Inc. v. Taamneh* [2023] that recommendation algorithms are merely part of the social media platform's infrastructure and declined to hold the platform liable for aiding and abetting terrorism. If mere automatic copying in response to users' actions did not rise to the level of causation, the even less predictable outputs of generative AI systems would seem to weigh against such a finding.

Second, the AI system, usually dictates the expressive aspects of AI-generated content, not the developer or user. The Copyright Office has determined that AI-generated content should not be copyrightable where the AI system, not a human, is generating the expressive content [Copyright Registration Guidance 2023]. A district court upheld this rule, noting the unpredictability of generative AI systems' outputs versus those of cameras, which generate mechanical reproductions "only after the photographer develops a 'mental conception' of the photograph" [Thaler v. Perlmutter 2023]. Unlike cameras, the user (and programmer) do not develop a specific "mental conception" before the work is created, just an idea expressed in a prompt. Therefore, the expressive content of the output—infringing or not—is being calculated by the AI system, not the human user or developer. The AI system selecting the expressive content means that its actions, rather than a human user's prompt, are likely causing the substantially similar, infringing output.

Third, holding developers or users liable for all infringements generated by an AI system would violate longstanding copyright law incentives-access norms and could have a chilling effect on AI innovation and competition. When applying or refining copyright law to accommodate new technologies, courts and Congress must be careful to avoid upsetting copyright's careful incentives-access balance. This is why Judge Whyte rejected direct liability as an appropriate claim for the Internet service provider in *Netcom* [1995], because it would effectively hold the provider liable for the mere provision of the product. A careless analysis could misbalance copyright and make developers or users bear the cost of any infringements despite their best efforts. Holding the developer or user liable would disincentivize new entrants in the generative AI industry [Abbott 2018]. Such a rule would create significant liability exposure ranging from $200 up to $150,000 per infringed work [Copyright Law of the United States 2010a]. While larger companies could continue, this risk of liability would likely limit the viability of new, smaller players entering the field, and could restrict the possible social benefits of AI.

Fourth, strict attribution of causation to developers or users would not fully serve tort law's goal of deterrence. It would subject those parties to liability despite the fact that they cannot effectively predict or prevent unforeseeable infringements due to the AI system's semiautonomous nature. Deterrence sometimes requires knowledge that one's acts would result in misconduct, but the semiautonomous nature of AI systems makes this unlikely because a user or developer would not know if or when infringement could occur [Lemley and Casey 2019]. Alternatively, deterrence could incentivize precautions, which requires the defendant to have been able to foresee the risk of injury and its magnitude in advance [Calo 2015]. Yet foreseeability is especially uncertain when events are complex or stochastic [Balganesh 2009]. It would be foolhardy to expect that *ex ante* testing is a panacea. No amount of testing can make a developer certain about what an AI system would do in response to a particular prompt, let alone an untested prompt. This issue is compounded by the fact that machine learning results in AI systems constantly improving and changing [Kroll et al. 2017].

4.2.4 Parallel Problems with a Principal-Agent Relationship. Making the AI system directly liable does not mean that the developer should be de facto liable for the AI system's infringements under a principal-agent relationship, as this would effectively impose the same de facto liability on developers without

the limit of causation. Agency law requires that a principal assents to the agent acting "on the principal's behalf and subject to the principal's control, and the agent manifests assent or otherwise consents so to act" [Restatement (Third) of Agency 2006]. The Supreme Court has described control as one of the "basic features of an agency relationship" [Hollingsworth v. Perry 2013]. Generally, agency takes the form of actual authority, apparent authority, or *respondeat superior*. Actual authority requires that "the agent reasonably believes, in accordance with the principal's manifestations to the agent, that the principal wishes the agent so to act." Apparent authority is when "a third party reasonably believes the actor has authority to act on behalf of the principal and that belief is traceable to the principal's manifestations." Finally, *respondeat superior* is the principle that "[a]n employer is subject to liability for torts committed by employees while acting within the scope of their employment." Underlying all three of these relationships is a consensual relationship between principal and agent and control by the principal [Restatement (Third) of Agency 2006].

A few scholars have previously proposed agency or *respondeat superior* as the optimal liability framework for AI systems. For example, Lior [2020], acknowledges that AI systems are neither tools nor humans and has advocated for a general AI *respondeat superior* doctrine, where AI systems are treated as the agents of their developers or human operators. Lior justifies this approach on the basis that AI systems are effectively judgment proof and cannot prevent the tortious behavior from occurring by themselves. Additionally, AI is expected to replace many human employees for whom their employers are currently liable under *respondeat superior*. Chopra and White [2011] similarly justify agency liability on the basis that offering the developer as the responsible party would seem to resolve the problem of compensating the injured party. In the copyright context, Chatterjee and Fromer [2019] recognized that even if a machine could be held directly liable, it would have no assets, rights, or legal status—effectively nullifying any practical benefits of such a ruling. As a practical matter, therefore, the AI system could potentially be treated as an agent of the developer or user who employs them.

These scholars correctly identify the problem of holding the AI system liable when it does not have assets to compensate injured parties, but agency law is a poor doctrinal and normative fit for generative AI. The relationship between the developer and generative AI system lack the consensual relationship and degree of control that are usually the *sine qua non* of agency law. There is no consensual relationship between an AI system developer and the AI system itself. While generative AI is becoming increasingly autonomous, it does not yet have the ability to manifest assent. Relatedly, an AI system seemingly could not perform the fiduciary duty of loyalty because it is a machine, not a human.

In addition, the level of control by a generative AI developer over the system is limited. The developer could terminate the system, but its control over specific outputs of the AI system are circumscribed by the AI's semiautonomous nature. The Supreme Court rejected an agency theory where the purported agents "decide for themselves, with no review, what arguments to make and how to make them" [Hollingsworth v. Perry 2015]. Similarly, generative AI creates unpredictable outputs without developer review at the time of creation. In the copyright context, the Ninth Circuit found that a principal-agent relationship could exist where there was "substantive supervision," the principal "selects and removes moderators on the basis of their performance," and the principal "control[s] the moderators' work schedule" [Mavrix Photographs, LLC v. Livejournal, Inc. 2017]. This level of control is lacking for generative AI, where there are no human employees and the creation of outputs occurs independently of individualized supervision by the developer.

Holding the developer liable as the principal of the AI agent is also undesirable from a normative perspective. The same reasons strict liability should be disfavored more broadly for artificial infringement apply here too. As described in the prior section (Section 4.2.3), strict liability in this context conflicts with copyright precedent on causation; the AI system is selecting the infringing expression, it could harm AI innovation and competition, and it would likely not properly advance deterrence but would unduly constrain AI's capabilities. In addition, de facto liability under an agency theory would remove the causation limitation, which, while limited, still helps balance copyright's competing goals of incentivizing creative works and providing public access to those works.

4.3 Secondary Liability for Humans

The proposal of making the AI system the direct infringer and eschewing a principal-agent relationship does not mean no human should be liable for artificial infringement. Holding the AI system directly liable for infringing outputs allows courts to ultimately anchor more nuanced and intent-sensitive (secondary) liability to a paying entity, such as the developer. Balkin [2015] recommended approaching legal doctrine from a pragmatic approach like this to accommodate new technologies, noting that "[w]hen we employ legal fictions of this kind, we substitute robot for human to allow the law to function effectively in the face of the legal enigmas posed by emergent behavior." If a court declined to hold the AI system liable, holding the user directly liable could achieve the same end of determining developer liability on a secondary basis. Lee et al. [2025], for example, previously suggested that developers could be held secondarily liable for users' direct infringements with AI systems. The distinction between direct and secondary liability—and the flexibility of secondary liability—has "long provided welcome breathing room for new technologies and uses" [Giblin and Ginsburg 2015].

Secondary liability is from common law and judges have crafted and refined the doctrine in response to technologies without the need for new laws. In my prior work [Goodyear 2025], I elucidated the polestar for refinements to secondary liability to maintain copyright's incentives-access balance: intent, or an actor's action or inaction, once aware of a particular alleged infringement, that is substantially certain to facilitate or further the infringement. By using intent as the balance polestar once again, courts and Congress can refine secondary liability to accommodate generative AI under copyright law. While vicarious liability is based on strict

liability, which is problematic for similar reasons to direct liability and agency, contributory liability and inducement offer a more flexible and intent-based framework.

4.3.1. Vicarious Liability. Courts impose vicarious copyright liability on an entity when it has the right and ability to control infringing content and receives a financial benefit directly attributable to said infringement [Perfect 10, Inc. v. Giganews, Inc. 2017]. Vicarious liability in copyright law descends from agency law and *respondeat superior* and, like them, imposes de facto liability based on control [Fonovisa, Inc. v. Cherry Auction,. Inc. 1996].

Vicarious liability for artificial infringement suffers from doctrinal and normative flaws. First, it is not clear that the developer could be vicariously liable under the doctrinal test. One of the two requirements for vicarious liability is having the right and ability to control the infringement. The definition of control is somewhat nebulous, but courts have started to coalesce around the ability to delete or remove online content and are reluctant to find it where there is less control [Business Casual Holdings, LLC v. YouTube, LLC 2023] [Luvdarts, LLC v. AT&T Mobility, LLC 2013]. A developer cannot predict, let alone remove generated outputs, suggesting that a court may not find the right and ability to control in the AI context. As Gervais [2020] has noted (albeit in the context of copyrightability), "the AI machine uses its own insights to create." Lee et al. [2025] have posited that AI system operators would have the right and ability to control because they could disable the entire system or filter the outputs. But requiring the disabling of the entire product goes much further than precedent has held [A&M Records, Inc. v. Napster, Inc. 2001] [Sony Music Entertainment v. Cox Communications, Inc. 2020] [Agence France Presse v. Morel 2013]. Online platforms who cannot prevent infringing content from appearing are typically not held to have the requisite control for vicarious liability [Routt v. Amazon.com, Inc. 2014] [Luvdarts, LLC v. AT&T Mobility, LLC 2013]. Likewise, here, a finding of requisite control would seem uncertain at best.

If a court did find control, that would trigger a second problem: vicarious infringement, like direct infringement and principal-agent relationships, would effectively impose de facto liability [Yen 2006]. Vicarious liability holds a party liable for the tortious actions of another based solely on their relationship without regard to fault [Cane 2002]. Therefore, it would hold the developer liable for any resulting infringements just like direct liability and agency law. This de facto—and potentially widespread—liability would lead to the same normative problems described above (Section 4.2.3), including conflicting with copyright precedent, ignoring causation as to the (infringing) expression, chilling innovation and competition, not achieving proper deterrence, and restricting the capabilities of AI.

Normatively, vicarious liability may be an even worse paradigm than direct liability for artificial infringement. The "volition or causation" requirement requires some degree of causation that could prevent a developer or user from being held directly liable, as explained above (Section 4.2.2). A vicarious party, however, is held liable for the direct infringer's infringements *regardless* of their causation. The lack of foreseeability still means that none of the stakeholders are able to fully assess the risks of the AI system and act appropriately in response, undermining deterrence. This has led some scholars to conclude that automated AI requires a paradigm shift away from strict liability and similar de facto rules such as vicarious liability altogether [Rachum-Twaig 2020] [Marchisio 2021] [Beckers and Teubner 2024].

4.3.2. Contributory Liability and Inducement. A novel but promising alternative use of holding the AI system directly liable is applying contributory liability and inducement for artificial infringement. Contributory copyright infringement liability requires knowing about the specific infringement and materially contributing to it [Perfect 10, Inc. v. Giganews, Inc. 2017]. Since generative AI has substantial noninfringing uses, knowledge of infringement cannot be imputed to the developer but must be determined based on specific infringements [Sony Corporation of America v. Universal City Studios, Inc. 1984]. Contributory liability is a fault-based standard that allows for more flexibility and more consideration of intent and tort law goals since it focuses on fault with regard to specific infringements [Yen 2006]. As I elucidated in my prior work [Goodyear 2025], Courts and Congress have crafted a contributory liability framework that usually considers platforms' actions in response to specific instances of infringements rather than infringement more generally because it focuses on platforms' specific intent. Unlike negligence, which looks to *ex ante* impositions on generative AI systems that could stifle innovation and do not necessarily consider intent, contributory liability scrutinizes parties' actions to determine if they acted to further the infringement once aware of it. Indeed, in *Sony* [1984], the Supreme Court barred holding a developer contributorily liable because it was generally foreseeable that its device could be used to infringe. Contributory liability therefore appears best situated to maintain balanced copyright in the face of the novel unpredictability of AI-generated outputs.

Once an AI system developer is aware that there is a specific infringement—whether through the rights owner or anyone else reporting it, or otherwise learning of it—it would be required to take a certain action in response to negate its material contribution to that infringement. Different courts apply different material contribution tests, including removing the infringement (i.e., notice-and-takedown, specifically the takedown component), providing the "site and facilities" for the infringement, or failing to take "simple measures" to prevent the infringement [A&M Records, Inc. v. Napster, Inc. 2001] [Perfect 10, Inc. v. Giganews, Inc. 2017]. Notice-and-takedown is perhaps the most prominent due to its parallel with the requirements under the Digital Millennium Copyright Act (DMCA), which provides a safe harbor for online service providers for the copyright infringements of their users in exchange for them meeting a series of requirements [Copyright Act of the United States 2010b].

Notice-and-takedown would not work for AI systems because the developer could not simply remove the infringing output once it has been reported. The generated work already exists in the world and was likely downloaded at that point. It may also be difficult to prevent a specific work from being infringed again. So-called

instance unlearning is a nascent approach, and although it has long-term promise, it currently incurs high computational cost and inaccuracies [Cooper et al. 2024]. It could even compromise the AI system's performance [Lee et al. 2025].

A mechanism that filters outputs that are too similar to a particular copyrighted work or are unlikely to be fair use could be another possibility. However, as prior scholarship has shown, mechanical determinations of fair use are far from perfect and can impede free speech [Burk 2019]. While a filter could be required, it is quite difficult to devise filters that are robust enough to capture more than mere surface-level matches but not so broad as to be proscriptively overinclusive of fair uses and other noninfringing outputs [Henderson et al. 2023b]. Indeed, in devising the DMCA system, Congress expressly did not require service providers to engage in proactive activities such as filtering for this reason [Copyright Act of the United States 2010b].

An alternative that builds upon notice-and-takedown while considering secondary liability's intent polestar for refining the law would be a more promising vehicle for maintaining copyright's incentives-access balance. Henderson et al. [2023a] noted the potential viability of a notice-and-takedown-like regime for AI-generated libel against public figures. The fact that there are several different material contribution tests allows some flexibility in creating a *sui generis* refinement to the contributory liability test rather than relying on one of the existing options.

A promising *sui generis* refinement option could be what I term "notice-and-revision." Instead of requiring takedowns, this structure would require expeditious *revision* of the model or tool to address the learned-of infringement. Revision could take a variety of forms, including filtering code, adjusting the model, adding training content to lower the chance of a similar output, or providing instructive feedback to the tool. Volokh [2023] proposed a similar notice-and-blocking structure for AI-generated defamation. Even if developers endeavor to take some of these actions, they are not part of the liability structure at present but merely ex post decisions to prevent similar infringements (and liability for those infringements) in the future. By interpreting notice-and-revision as the material contribution requirement for contributory liability, developers avoid liability for unforeseeable infringements in the first instance and can then revise the system to prevent similar infringements in the future. While the revision will not necessarily prevent all future infringement, even of the same copyrighted work, it puts the onus on the AI system developer to improve the system once they are away of its specific infringement capabilities. As Volokh [2023] noted in the defamation context, once the company is on notice, "[p]resumably the company could then add [post-processing content filtering] code that would prevent these particular allegations [or, in this case, infringements] . . . from being output." Indeed, this is seemingly what happened with ChatGPT, which fixed their "Snoopy" problem once Sag identified it or Open AI otherwise became aware of it [Sag 2024]. Thus, notice-and-revision offers a balanced approach that places certain reasonable obligations on the developer only once they are aware of the problem—holding the developer liable where they have intentionally acted or not acted in response to knowledge of specific infringements.

Notice-and-revision provides at least four benefits that help maintain the incentives-access balance: (1) it applies *ex post* restrictions, (2) it accounts for the unpredictability of AI-generated outputs, (3) it targets the most harmful uses of infringing outputs, and (4) it does not preclude courts or Congress imposing additional *ex ante* industry best practices on developers.

First, direct, agency, or vicarious liability would impose *ex ante* restrictions on AI—the developer would be held liable for any infringements that occur. But, as explained above (Section 4.2.3), this categorical liability rule would create a significant chilling effect on AI innovation for all but the wealthiest and most risk-tolerant companies. Notice-and-revision, however, is more akin to an *ex post* restriction where liability would only be triggered once the developer has *specific* knowledge of the infringement problem and fails to act. General knowledge that an AI system could be used for infringement is insufficient for contributory liability so long as there are substantial noninfringing uses (which should typically be the case with AI systems) [Sony Corporation of America v. Universal City Studios, Inc. 1984].

Second, to realize the benefits of AI, we must accept that AI-generated outputs will sometimes be unpredictable and account for that in crafting liability. But direct, principal-agent, and vicarious liability fail to do this because they would always hold the developer or user liable. This was one of the reasons Rachum-Twaig [2020] found extant law inadequate for addressing AI harms. Instead, Rachum-Twaig proposed supplemental liability safe harbors in exchange for meeting certain requirements, including ongoing support duties and monitoring. Contributory liability already incorporates ongoing support duties in the sense that the developer must respond to reported infringement or risk being held liable [Perfect 10, Inc. v. Amazon.com, Inc. 2007]. It does not incorporate monitoring and, more broadly, copyright law has typically placed the duty to monitor on the rights owner [Copyright Act of the United States 2010b]. The previously mentioned filter for copyrighted content could be a possible solution, but it would still have its inherent issue of cutting too broadly to capture only infringement and not lawful free expression. Contributory liability accounts for the unpredictable nature of AI-generated outputs and intent by looking to fault rather than imposing a static, stringent test. The notice-and-revision structure would require that the developer have knowledge of the specific infringement that is occurring and then revise the algorithms to mitigate this danger. The developer's action or inaction provides indicia of their intent. If the developer fails to revise the model, they would be liable for infringement. This would require specific intent to not act knowing that infringement could likely recur, rather than de facto liability for the unforeseeable infringements of an AI system.

Third, the difficulty in identifying infringing outputs may be advantageous for society because only the most harmful uses would be implicated. While AI systems generate millions of outputs, rights owners and others may only be aware of specific infringements in a fairly small subset of circumstances. Those cases are likely to be instances when the output is being used in a

prominent and public way, such as for commercial purposes. These are the sort of uses that pose the greatest threat to the copyright owners' own rights, as demonstrated, for example, by the importance of commerciality and market effect in copyright law's fair use analysis. This structure would allow many personal, noncommercial uses to go undetected. These pose less danger to rights owners and, indeed, similar uses outside of the AI context go undiscovered and unlitigated. Rights owners and courts have even been sympathetic to these types of uses, as shown in *Cablevision* [2008] and the Supreme Court's earlier *Sony* [1984] decision, where individuals who copied televised content for their own personal enjoyment were not parties to the litigation even though they were the direct infringers.

Finally, notice-and-revision for contributory liability does not preclude courts or Congress from crafting additional requirements for AI companies to comply with industry best standards to lower the amount of infringements. Studies have shown that AI systems have different amounts of infringements, ranging from OpenAI's 44% copyright detection to Meta's 8% [CopyrightCatcher 2024]. Other scholars such as Henderson et al. [2023b] have discussed potential technical *ex ante* measures developers could take. While it is beyond the scope of this Article, developers should consider implementing these best practices to the extent they help reduce infringements and improve the novelty of AI systems' outputs. Even if developers do not implement them, they would still have to undertake best efforts to improve their system once they are aware of the specific infringement problem. Notice-and-revision only provides a one-time safe harbor per infringement issue rather than an excuse to not comply with industry practices.

Beyond the notice-and-revision structure for contributory liability, the inducement doctrine could also be used to capture more nefarious actors that could also qualify as causes of infringement subject to direct liability if the AI system were not held liable. While the AI system could be the direct infringer in all cases, as contemplated above (Section 4.2.2), intentionally infringing developers and users would still be liable under the inducement doctrine. As established by the Supreme Court in *Metro-Goldwyn-Mayer Studios Inc. v. Grokster, Ltd.* [2005], one is liable for infringement if they were "actively inducing" it through "purposeful, culpable expression and conduct." Therefore, a developer that intends their AI system to be an infringement generation device or a user who actively seeks to have the AI system infringe could be held liable for inducement.

5 CONCLUSION

When artificial infringement litigation advances past fair use considerations this interpretation of how the law should apply will hopefully serve as a useful reference for policymakers and courts, as well as computer scientists. As courts and Congress address ever more novel and complex technologies, they must use all the tools at their disposal to maintain copyright's incentives-access balance. Refinements to direct liability are an underconsidered possibility that previously led to the emergence of the "volition or causation" requirement in mechanical infringement cases. While it is a departure from the pure direct liability of corporeal and mechanical infringement, the imposition of direct liability on the AI system is grounded in the causation polestar courts and Congress have previously used. It also operates in tandem with secondary liability refinements and fair use to, together, maintain copyright's incentives-access balance despite the acute challenges of artificial infringement.

REFERENCES

A&M Records, Inc. v. Napster, Inc. 2001. 239 F.3d 1004 (9th Cir. 2001).

Ryan Abbott. 2018. The Reasonable Computer: Disrupting the Paradigm of Tort Liability. *George Washington Law Review* 86 (2018), 1.

Ryan Abbott and Alex Sarch. 2019. Punishing Artificial Intelligence: Legal Fiction or Science Fiction. *UC Davis Law Review* 53 (2019), 323.

ABKCO Music, Inc. v. Sagan 2022. 50 F.4th 309 (2d Cir. 2022).

Kenneth S. Abraham and Robert L. Rabin. 2019. Automated Vehicles and Manufacturer Responsibility for Accidents: A New Legal Regime for a New Era. *Virginia Law Review* 105 (2019), 127.

Agence France Presse v. Morel 2013. 934 F. Supp. 2d 547 (S.D.N.Y. 2013).

American Broadcasting Cos. v. Aereo, Inc. 2014. 573 U.S. 431 (2014).

BJ Ard. 2025. Copyright's Latent Space: Generative AI and the Limits of Fair Use. *Cornell Law Review* 110 (2025). Forthcoming.

Author's Guild v. Google, Inc. 2015. 804 F.3d 202 (2d Cir. 2015).

Shyamkrishna Balganesh. 2009. Foreseeability and Copyright Incentives. *Harvard Law Review* 122 (2009), 1569.

Jack M. Balkin. 2015. The Path of Robotics Law. *California Law Review Circuit* 6 (2015), 45.

Solon Barocas and Andrew D. Selbst. 2016. Big Data's Disparate Impact. *California Law Review* 104 (2016), 671.

Anna Beckers and Gunther Teubner. 2023. Responsibility for Algorithmic Misconduct: Unity or Fragmentation of Liability Regimes? *Yale Journal of Law and Technology* 25 (2023), 76.

Boisson v. Banian, Ltd. 2001. 273 F.3d 262 (2d Cir. 2001).

Samuel R. Bowman. 2023. Eight Things to Know About Large Language Models. arXiv:2304.00612. Retrieved from https://arxiv.org/pdf/2304.00612.pdf.

Oren Bracha. 2023. The Work of Copyright in the Age of Machine Production. https://papers.ssrn.com/sol3/papers.cfm?abstract_id=4581738.

Brouse v. United States 1949. 83 F. Supp. 373 (N.D. Ohio 1949).

Dan L. Burk. 2019. Algorithmic Fair Use. *University of Chicago Law Review* 86 (2019), 283.

Business Casual Holdings, LLC v. YouTube, LLC 2023. No. 22-3007-cv, 2023 WL 6842449 (2d Cir. Oct. 17, 2023).

BWP Media USA, Inc. v. T & S Software Associates, Inc. 2017. 852 F.3d 436 (5th Cir. 2017).

Ryan Calo. 2015. Robotics and the Lessons of Cyberlaw. *California Law Review* 103 (2015), 513.

Ryan Calo, Ivan Evtimov, Earlence Fernandes, Tadayoshi Kohno, and David O'Hair. 2018. Is Tricking a Robot Hacking? https://papers.ssrn.com/sol3/papers.cfm?abstract_id=3150530.

Came v. Micou 2005. No. 04-CV-1207, 2005 WL 1500978 (M.D. Pa. June 23, 2005).

Peter Cane. 2002. *Responsibility in Law and Morality*. Hart Publishing, Portland, OR.

Capitol Records, LLC v. ReDigi Inc. 2013. 934 F. Supp. 2d 640 (S.D.N.Y. 2013).

Nicholas Carlini et al. 2023. Extracting Training Data from Diffusion Models. arXiv:2301.13188. Retrieved from https://arxiv.org/pdf/2301.13188.pdf.

Cartoon New Tork, LP, LLLP v. CSC Holdings, Inc. 2008. 536 F.3d 121 (2d Cir. 2008).

Mala Chatterjee and Jeanne C. Fromer. 2019. Minds, Machines, and the Law: The Case of Volition in Copyright Law. *Columbia Law Review* 119 (2019), 1887.

Samir Chopra & Laurence F. White. 2011. Tort Liability for Artificial Agents. In *A Legal Theory for Autonomous Artificial Agents* (2011). University of Michigan Press, 119–52.

Comcast Corp. v. National Association of African American-Owned Media. 2020. 589 U.S. 327 (2020).

A. Feder Cooper, Jonathan Frankle, and Christopher De Sa. 2022. Non-Determinism and the Lawlessness of Machine Learning Code. In *Proceedings of the 2022 Symposium on Computer Science and Law (CSLAW '22)*, November 1–2, 2022, Washington, DC, USA. ACM, New York, NY, USA, 8 pages. https://doi.org/10.1145/3511265.3550446.

A. Feder Cooper et al. 2024. Machine Unlearning Doesn't Do What You Think: Lessons for Generative AI Policy, Research, and Practice. arXiv:2412.06966. Retrieved from https://arxiv.org/abs/2412.06966.

A. Feder Cooper and James Grimmelmann. The Files Are in the Computer: On Copyright, Memorization, and Generative AI. 98 *Chicago-Kent Law Review* 98 (2025). Forthcoming.

Constitution of the United States. 1789. https://www.law.cornell.edu/constitution.
Copyright Law of the United States 1990. https://www.law.cornell.edu/uscode/text/17/102 U.S.C. 17, 102.
Copyright Law of the United States 1992. https://www.law.cornell.edu/uscode/text/17/107 U.S.C. 17, 107.
Copyright Law of the United States 2002. https://www.law.cornell.edu/uscode/text/17/106 U.S.C. 17, 102.
Copyright Law of the United States 2010a. https://www.law.cornell.edu/uscode/text/17/504 U.S.C. 17, 504.
Copyright Law of the United States 2010b. https://www.law.cornell.edu/uscode/text/17/512 U.S.C. 17, 512.
Copyright Law of the United States 2019. https://www.law.cornell.edu/uscode/text/17/501 U.S.C. 17, 501.
Copyright Registration Guidance: Works Containing Material Generated by Artificial Intelligence. Federal Register 88 (Mar. 16, 2023), 16190.
Corporations Law of Wyoming 2023. https://law.justia.com/codes/wyoming/title-17/chapter-31/article-1/section-17-31-101/ Wyoming Statute 17-31-101.
Antonio Davola. 2018. A Model for Tort Liability in a World of Driverless Cars: Establishing a Framework for the Upcoming Technology. Idaho Law Review 54 (2018), 591.
Robert C. Denicola. 2016. Volition and Copyright Infringement. *Cardozo Law Review* 37 (2016), 1259.
Deven R. Desai and Joshua A. Kroll. 2017. Trust But Verify: A Guide to Algorithms and the Law. *Harvard Journal of Law and Technology* 31 (2017), 1.
Disney Enterprises, Inc. v. Hotfile Corp. 2011. 798 F. Supp. 2d 1303 (S.D. Fla. 2011).
Mindy Nunez Duffourc. 2023. Malpractice by the Autonomous AI Physician. *Journal of Law, Technology and Policy* (2023), 1.
Field v. Google Inc. 2006. 412 F. Supp. 2d 1106 (D. Nev. 2006).
Fonovisa, Inc. v. Cherry Auction, Inc. 1996. 76 F.3d 259 (9th Cir. 1996).
Fox Broadcasting Co. v. Dish Network LLC 2014. 747 F.3d 1060 (9th Cir. 2014).
Deep Ganguli, Danny Hernandez, Liane Lovitt, Nova Dassarma, Tom Henighan, Andy Jones, Nicholas Joseph, Jackson Kernion, Ben Mann, Amanda Askell, Yuntao Bai, Anna Chen, Tom Conerly, Dawn Drain, Nelson Elhage, Sheer El Showk, Stanislav Fort, Zac Hatfield-Dodds, Scott Johnston, Shauna Kravec, Neel Nanda, Kamal Ndousse, Catherine Olsson, Daniela Amodei, Tom Brown, Jared Kaplan, Sam McCandlish, Chris Olah, Dario Amodei, and Jack Clark. 2022. Predictability and Surprise in Large Generative Models. In *2022 ACM Conference on Fairness, Accountability, and Transparency (FAccT '22), June 21–24, 2022, Seoul, Republic of Korea*. ACM, New York, NY, USA, 18 pages. https://doi.org/10.1145/3531146.3533229.
Sara Gerke. 2021. Health AI for Good Rather Than Evil? The Need for a New Regulatory Framework for AI-Based Medical Devices. *Yale Journal of Health Policy, Law, and Ethics* 20 (2021), 433.
Daniel J. Gervais. 2020. The Machine as Author. *Iowa Law Review* 105 (2020), 2053.
Rebecca Giblin and Jane C. Ginsburg. 2015. We (Still) Need to Talk About Aereo: New Controversies and Unresolved Questions After the Supreme Court's Decision. *Columbia Journal of Law and the Arts* 38 (2015), 109.
Jane C. Ginsburg. 2002. How Copyright Got a Bad Name for Itself. *Columbia Journal of Law and the Arts* 26 (2002), 61.
Michael P. Goodyear. 2025. Infringing Information Architectures. *UC Davis Law Review* 58 (2025). Forthcoming.
Patrick R. Goold. 2015. Is Copyright Infringement a Strict Liability Tort? *Berkeley Technology Law Journal* 30 (2015), 305.
Andres Guadamuz. 2024. A Scanner Darkly: Copyright Liability and Exceptions in Artificial Intelligence Inputs and Outputs. *GRUR International* 73 (2024), 111.
Peter Henderson, Tatsunori Hashimoto, and Mark Lemley. 2023a. Where's the Liability in Harmful AI Speech? *Journal of Free Speech Law* 3 (2023), 589.
Peter Henderson, Xuechen Li, Dan Jurafsky, Tatsunori Hashimoto, Mark A. Lemley, and Percy Liang. 2023b. Foundation Models and Fair Use. *Journal of Machine Learning Research* 23 (2023), 1.
Hollingsworth v. Perry 2013. 570 U.S. 693 (2013).
F. Patrick Hubbard. 2014. "Sophisticated Robots": Balancing Liability, Regulation, and Innovation. *Florida Law Review* 66 (2014), 1803.
P. Bernt Hugenholtz and João Pedro Quintais. 2021. Copyright and Artificial Creation: Does EU Copyright Law Protect AI-Assisted Output? *IIC – International Review of Intellectual Property and Competition Law* 52 (2021), 1190.
Hunley v. Instagram, LLC 2023. 73 F.4th 1060 (9th Cir. 2023).
Margot E. Kaminski. 2023. Regulating the Risks of AI. *Boston University Law Review* 103 (2023), 1347.
Lewis A. Kaplan. 2000. Keynote Address: Resolving Tensions Between Copyright and the Internet. *American University Law Review* 50 (2000), 409.
Karney v. Leonard Transportation Corp. 2008. 561 F. Supp. 2d 260 (D. Conn. 2008).
Curtis E.A. Karnow. 1996. Liability for Distributed Artificial Intelligences. *Berkeley Technology Law Journal* 11 (1996), 147.
Curtis E. A. Karnow. 2015. The Application of Traditional Tort Theory to Embodied Machine Intelligence. In *Robot Law*, Ryan Calo, A. Michael Froomkin, and Ian Kerr (Eds.). Elgar, 51–77.
Kelly v. Arriba Soft Corp. 2003. 336 F.3d 811 (9th Cir. 2003).
Joshua A. Kroll et al. 2017. Accountable Algorithms. *University of Pennsylvania Law Review* 165 (2017), 633.
Katherine Lee, A. Feder Cooper, and James Grimmelmann. 2025. Talkin' 'Bout AI Generation: Copyright and the Generative-AI Supply Chain. *Journal of the Copyright Society* (2025). Forthcoming.
Timothy B. Lee and James Grimmelmann. Why the New York Times Might Win Its Copyright Lawsuit Against OpenAI. *Ars Technica* (20 February 2024). https://arstechnica.com/tech-policy/2024/02/why-the-new-york-times-might-win-its-copåyright-lawsuit-against-openai.
Mark Lemley and Bryan Casey. 2019. Remedies for Robots. *University of Chicago Law Review* 86 (2019), 1311.
Mark Lemley and Bryan Casey. 2021. Fair Learning. *Texas Law Review* 99 (2021), 744.
Amanda Levendowski. 2018. How Copyright Law Can Fix Artificial Intelligence's Implicit Bias Problem. *Washington Law Review* 93 (2018), 579.
Lexmark International, Inc. v. Static Control Components. 2014. 572 U.S. 118 (2014).
Anat Lior. 2020. AI Entities as AI Agents: Artificial Intelligence Liability and the AI Respondeat Superior Analogy. *Mitchell Hamline Law Review* 46 (2020), 1043.
Lydia Pallas Loren. 2003. Untangling the Web of Music Copyrights. *Case Western Reserve Law Review* 52 (2003), 673.
Luvdarts, LLC v. AT&T Mobility, LLC 2013. 710 F.3d 1068 (9th Cir. 2013).
MAI System Corp. v. Peak Computer, Inc. 1993. 991 F.2d 511 (9th Cir. 1993).
Emiliano Marchisio. 2021. In Support of "No-Fault" Civil Liability Rules for Artificial Intelligence. 1 *SN Social Sciences* (2021), 1.
Mavrix Photographs, LLC v. Livejournal, Inc. 2017. 873 F.3d 1045 (9th Cir. 2017).
Metro-Goldwyn-Mayer Studios Inc. v. Grokster, Ltd. 2005. 545 U.S. 913 (2005).
Paul B. Miller. 2021. Corporate Personality, Purpose, and Liability. In *Research Handbook on Corporate Purpose and Personhood*, Elizabeth Pollman and Robert B. Thompson (Eds.). Elgar, 222–39.
Christina Mulligan. 2018. Revenge Against Robots. *South Carolina Law Review* 69 (2018), 579.
David Nimmer. 2019. Volition in Violation of Copyright. *Columbia Journal of Law and the Arts* 43 (2019), 1.
Parker v. Google, Inc. 2006. 422 F. Supp. 2d 492 (E.D. Pa. 2006).
Patronus AI. 2024. CopyrightCatcher, the First Copyright Detection API for LLMs. https://www.patronus.ai/blog/introducing-copyright-catcher.
Perfect 10, Inc. v. Amazon.com, Inc. 2007. 508 F.3d 1146 (9th Cir. 2007).
Perfect 10, Inc. v. Giganews, Inc. 2017. 847 F.3d 657 (9th Cir. 2017).
Petroleos Mexicanos Refinacion v. M/T KING A. 2009. 554 F.3d 99 (3d Cir. 2009).
W. Nicholson Price II and I. Glenn Cohen. 2024. Locating Liability for Medical AI. *DePaul Law Review* 73 (2024), 339.
Omri Rachum-Twaig. 2020. Whose Robot Is It Anyway?: Liability for Artificial-Intelligence-Based Robots. *University of Illinois Law Review* (2020), 1141.
Religious Technology Center v. Netcom On-Line Communication Services, Inc. 1995. 907 F. Supp. 1361 (N.D. Cal. 1995).
Rentmeester v. Nike, Inc. 2018. 883 F.3d 1111 (9th Cir. 2018).
Restatement (Third) of Agency. 2006. ALI.
Eleonora Rosati. 2024. Infringing AI: Liability for AI-Generated Outputs Under International, EU, and UK Copyright Law. *European Journal of Risk Regulation* (2024), 1.
Routt v. Amazon.com, Inc. 2014. 584 F. App'x 713 (9th Cir. 2014).
Matthew Sag. 2023. Copyright Safety for Generative AI. *Houston Law Review* 61 (2023), 295.
Matthew Sag (@matthewsag). *X* (21 February 2024). https://twitter.com/matthewsag/status/1760429736344383671.
Pamela Samuelson et al. 2010. The Copyright Principles Project: Directions for Reform. *Berkeley Technology Law Journal* 25 (2010), 1175.
Pamela Samuelson. 2023. Generative AI Meets Copyright. *Science* 381 (2023), 158.
Pamela Samuelson. 2024. Fair Use Defenses in Disruptive Technology Cases. *UCLA Law Review* 71 (2024), 1484.
Matthew U. Scherer. 2018. Of Wild Beats and Digital Analogues: The Legal Status of Autonomous Systems. *Nevada Law Journal* 19 (2018), 259.
Andrew Selbst. 2019. Negligence and AI's Human Users. *Boston University Law Review* 100 (2019), 1315.
Selle v. Gibb 1984. 741 F.2d 896 (7th Cir. 1984).
Jatinder Singh, Ian Walden, Jon Crowcroft, and Jean Bacon. 2016. Responsibility & Machine Learning: Part of a Process. https://papers.ssrn.com/sol3/papers.cfm?abstract_id=2860048.
Sony Corporation of America v. Universal City Studios, Inc. 1984. 464 U.S. 417 (1984).
Sony Music Entertainment v. Cox Communications, Inc. 2020. 464 F. Supp. 3d 795 (E.D. Va. 2020).
Stewart v. Abend 1990. 495 U.S. 207 (1990).
Thaler v. Perlmutter. 2023. 687 F. Supp. 3d 140 (D.D.C. 2023).
Three Boys Music Corp. v. Bolton 2000. 212 F.3d 477 (9th Cir. 2000).
Charlotte A. Tschider. 2021. Medical Device Artificial Intelligence: The New Tort Frontier. *BYU Law Review* 46 (2021), 1551.
Twitter, Inc. v. Taamneh 2023. 143 S. Ct. 1206 (2023).

A.V. ex rel. Vanderhye v. iParadigms, LLC 2009. 562 F.3d 630 (4th Cir. 2009).
Jacob Victor. 2020. Reconceptualizing Compulsory Copyright Licenses. *Stanford Law Review* 72 (2020), 915.
John Villasenor. 2014. Products Liability and Driverless Cars: Issues and Guiding Principles for Legislation. https://www.brookings.edu/wp-content/uploads/2016/06/Products_Liability_and_Driverless_Cars.pdf.
David C. Vladeck. 2014. Machines Without Principles: Liability Rules and Artificial Intelligence. *Washington Law Review* 89 (2014), 117.
Eugene Volokh. 2023. Large Libel Models? Liability for AI Output. *Journal of Free Speech Law* 3 (2023), 489.
Alfred C. Yen. 2006. Third-Party Copyright Liability After Grokster. *Minnesota Law Review* 91 (2006), 184.